The dynamical characterisation of spinning particles around a short-hair black hole

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Outline of this presentation:--

- Introduction to our spacetime of interest.
- Spinning Particles in curved spacetime.
- Methodology used
- Results
- Applications
- Conclusion
- Limitations and Future directions
- Some References

Introduction:--

- Meaning of hair in the context of Black Holes (Bhs).
- No-short-hair theorem.
- Metric of our Spacetime,

$$ds^2 = -f(r)dt^2 + \frac{1}{f(r)}dr^2 + r^2(d\theta^2 + sin^2\theta d\phi^2)$$

Where,

$$f(r) = 1 - \frac{2M}{r} + \frac{Q_m^{2k}}{r^{2k}}.$$

- **M $\rightarrow$ mass of BH,**
- **Q_m** $\rightarrow$ strength of the BH's hair,
- k $\rightarrow$ parameter deciding various cases of BH,
  - k = 1, Reissner-Nordström BH,
    k = 1.5, short hair Bhs,
    k = 2, short hair BH which behaves like quantum corrected BH.

Spinning particles in curved spacetimes

- Non spinning particles follow geodesics.
- Spin deviates from geodesics.
- Why to study spinning particles ?
- Why to study spinning particles in curved spacetimes?
- Why do we need different formalism for this ?
- Why this study is done under approximations ?

Methodology Used:--

- Conserved quantities

$$\mathcal{Q} = \mathcal{K}^{\mu} P_{\mu} - \frac{1}{2} S^{\mu\nu} \mathcal{K}_{\mu;\nu}.$$

- Mathison-Papapetrou-Dixon (MPD) equations

$$\frac{DP^{\mu}}{D\lambda} = -\frac{1}{2} R^{\mu}_{\nu\alpha\beta} u^{\nu} S^{\alpha\beta},$$
$$\frac{DS^{\mu\nu}}{D\lambda} = P^{\mu} u^{\nu} - P^{\nu} u^{\mu},$$

- Effective potential

$$\frac{(P^r)^2}{m^2} = A\bar{e}^2 + B\bar{e} + C$$

$$V_{\text{eff}} = \frac{-B + \sqrt{B^2 - 4AC}}{2A}.$$

- Tulczyjew spin-supplementary condition

$$P_{\mu} S^{\mu\nu} = 0.$$

Mass shell condition

$$m^2 = -P^{\mu} P_{\mu},$$

spin magnitude constraint

$$s^2 = \frac{1}{2} S^{\mu\nu} S_{\mu\nu}.$$

- q value

$$q = \frac{\omega_{\phi}}{\omega_r} - 1 = \frac{\Delta\phi}{2\pi} - 1 = w + \frac{v}{z}.$$

Where,

$$\Delta\phi = \oint d\phi = 2\int_{\phi_1}^{\phi_2} d\phi = 2\int_{r_1}^{r_2} \frac{\dot{\phi}}{\dot{r}} dr.$$

Results:--

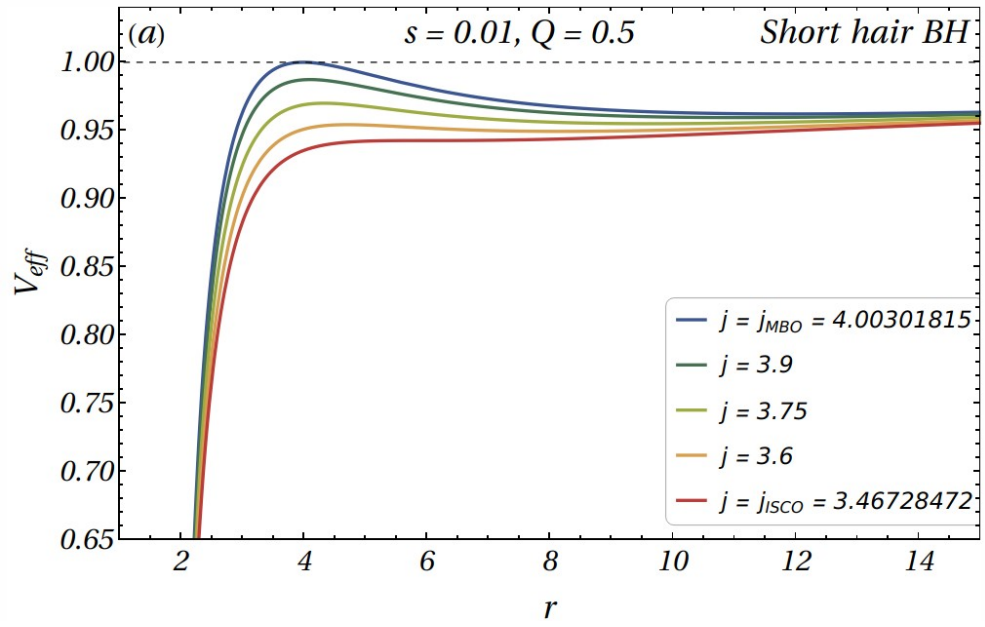
- **Innermost Stable Circular Orbit (ISCO)**

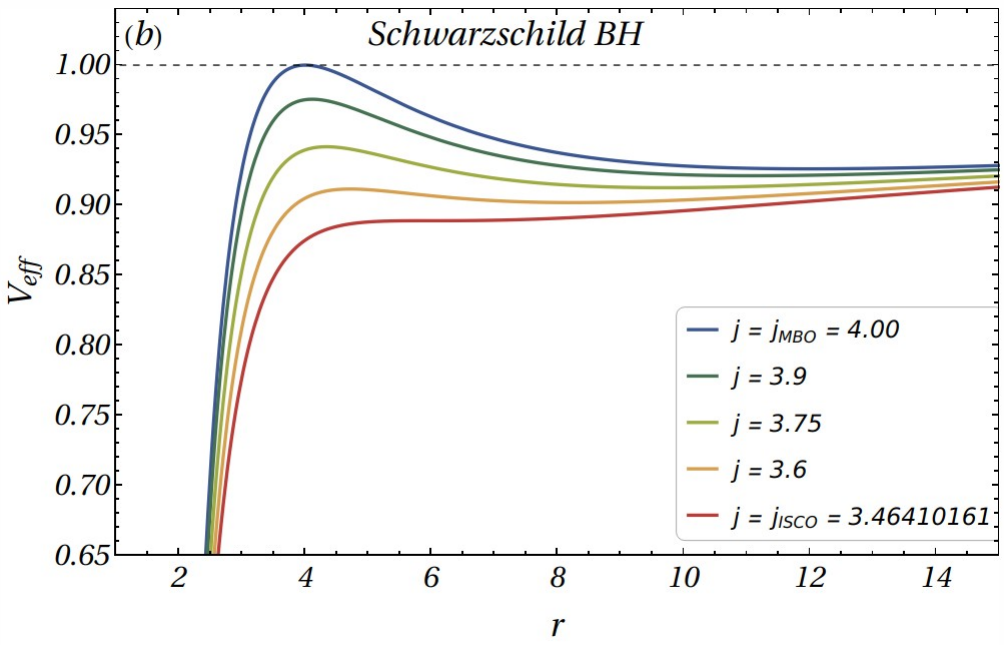
s	Q	r_{ISCO}	L_{ISCO}	E_{ISCO}
0	0	6.0000000	3.4641016	0.97098354
0	0.2	5.9997555	3.4640631	0.970983
0	0.5	5.9904177	3.4625938	0.970948
0	0.8	5.9359217	3.4540582	0.970747
0.01	0	5.9836297	3.4688012	0.970901
0.01	0.2	5.9833846	3.4687625	0.9709
0.01	0.5	5.9740246	3.4672847	0.970865
0.01	0.8	5.9193951	3.4586995	0.970662
0.1	0	5.8325385	3.5097335	0.970113
0.1	0.2	5.8322882	3.5096926	0.970112
0.1	0.5	5.8227254	3.5081345	0.970074
0.1	0.8	5.7668714	3.4990739	0.969853
0.2	0	5.6561732	3.5520576	0.96914
0.2	0.2	5.6559169	3.5520141	0.969139
0.2	0.5	5.6461244	3.5503551	0.969097
0.2	0.8	5.5888830	3.5406984	0.968852

- **Marginally Bound Orbit (MBO)**

s	Q	r_{MBO}	j_{MBO}	E_{MBO}
0	0	4.0000000	4.0000000	1
0	0.2	3.9998000	3.9999500	1
0	0.5	3.9921500	3.9980397	1
0	0.8	3.9471108	3.9868789	1
0.01	0	3.9899765	4.0049859	1
0.01	0.2	3.9897763	4.0049357	1
0.01	0.5	3.9821192	4.0030182	1
0.01	0.8	3.9370351	3.9918144	1
0.1	0	3.8976095	4.0485365	1
0.1	0.2	3.8974078	4.0484846	1
0.1	0.5	3.8896941	4.0464997	1
0.1	0.8	3.8442558	4.0348957	1
0.2	0	3.7902544	4.0938948	1
0.2	0.2	3.7900517	4.0938408	1
0.2	0.5	3.7822987	4.0917764	1
0.2	0.8	3.7366140	4.0796995	1

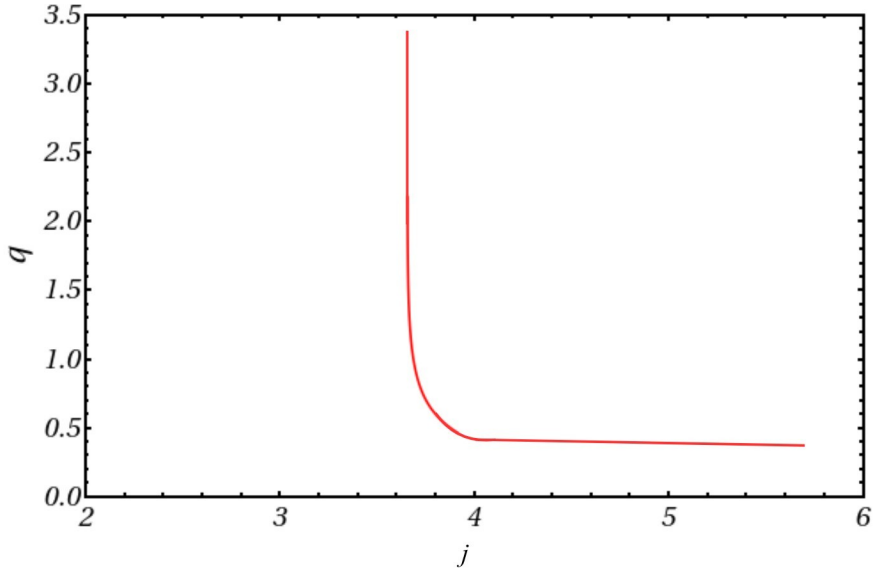
Effective potential

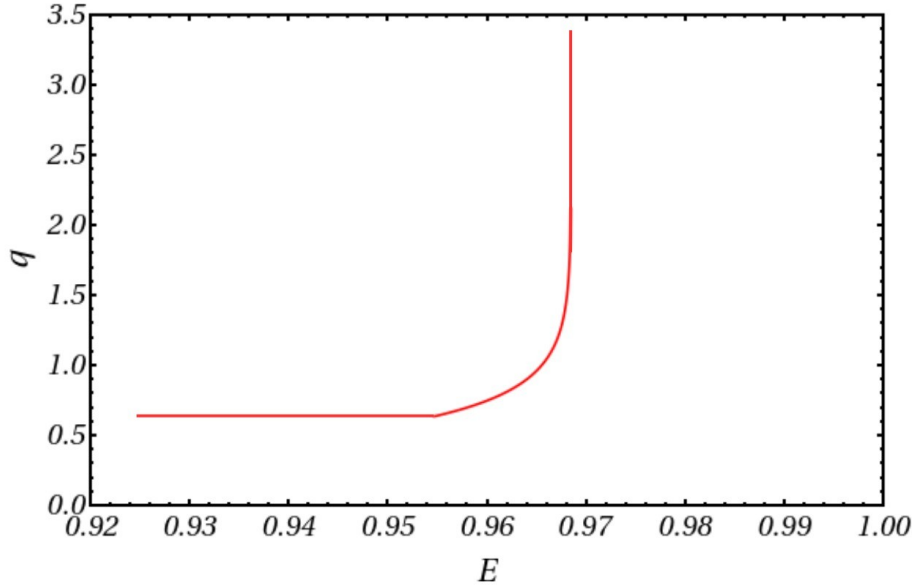




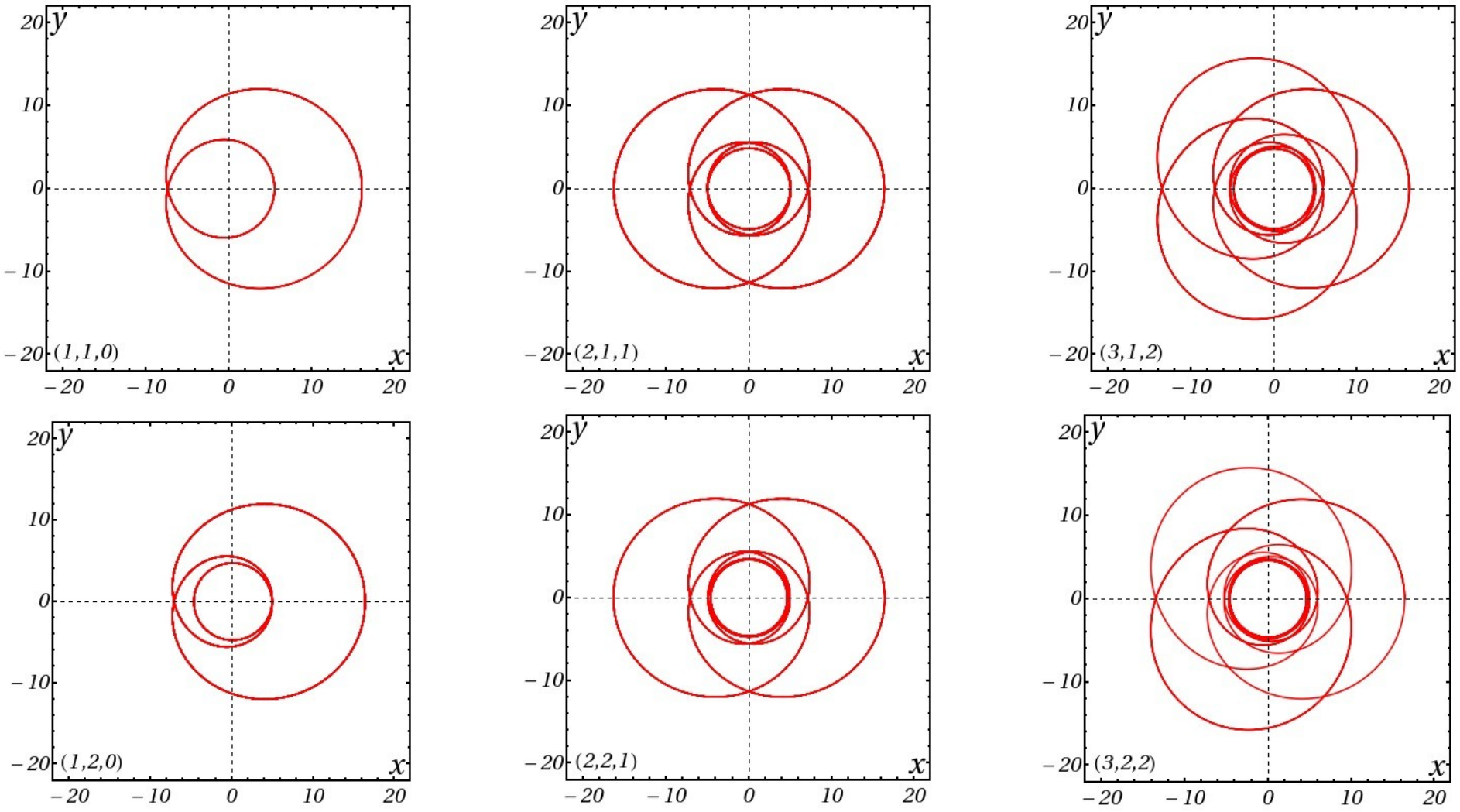
q-value

(s = 0.01, Q = 0.5)





<u>Periodic orbits</u> (s = 0.01, Q = 0.5)



Applications of present work:--

- Accretion disk physics: Spin-modified ISCO affects inner disk structure and emission spectra.
- Gravitational wave modeling : Spin–curvature coupling alters orbital dynamics in extreme mass ratio inspirals (EMRIs).
- Testing non-standard black holes : Short-hair parameter Q_m introduces deviations from Schwarzschild, offering a probe of beyond-GR effects.
- Astrophysical realism : Incorporating spin improves modelling of realistic particle motion in strong gravity.
- Long term detection for observable quantities : Periodic orbits gives long term detection possibilities for observable phenomena.

Conclusion:--

- Short hair BH which circumvents the “no-short-hair” theorem.
- We used MPD equations along with the Tulczyjew supplementary condition for this analysis.
- Both spin s and hair parameter Q_m significantly modify: ISCO, MBO, Effective potential and hence the overall orbital structure.
- Deviations increase with higher sand Q_m.
- Schwarzschild limit is recovered for $s \to 0$, $Qm \to 0$.

Limitations and Future directions:--

Limitations

- We approximated linear order in spin.
- No backreaction is considered, and the study is restricted to equatorial motion.
- The background is considered to be static, spherically symmetric and non- rotating.

Future directions

- Can be Extended to Kerr or rotating spacetimes.
- The higher order spin corrections can be implimented.
- The periodic orbits can be used to model gravitational wave signatures.
- non-equatorial and chaotic motion can be explored.

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Thank You